Web Analytics

MSIS 521
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FOSTER SCHOOL OF BUSINESS
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Objectives

- Web analytics for own website
  - What customers look at
  - Where they come from
  - How to engage them
- Web analytics for monitoring competitors
  - How customers behave in general
  - Why they go to competitors
  - What they do there

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Data-Driven Decision Making

- Common scenario:
  - Boss, designer, employee “knows what works best” (for them).
  - Boss, designer, employee wants to do site design.
- Common error: Think that we know what customers want.
- 80% of the time we are wrong about what a customer wants or expects from site experience.
- The only truth: Constant experimentation and testing improves customer “relevancy” and improves conversion.
- The measurement mindset: make decisions based on data (transactional data and experiments).

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What Data?

How customers behave in our own website

- Micro / In-site / Quantitative
  - Eyetracking
  - Clickstreams, log analysis
- Meso / In-site / Qualitative
  - Surveys
  - Lab-usability tests
  - In-site tests

How customers behave in other websites

- Macro / Across-site
  - Panel data (ComScore, Alexa, SimilarWeb)
  - Search engine data (Google Trends)

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Numbers Are Important!

The economic effect of a 1% increase in conversion rate

Measure	Symbol	Calculation	Before	After
Visitors	v		100,000	100,000
Cost per visit	c		1.00	1.00
Cost of all visits	c_T	$v \times c$	$100,000	$100,000
Conversion rate	r		3.0%	4.0%
Conversions	C	$r \times v$	3,000	4,000
Revenue per conversion	V		$75	$75
Total revenue	T	$V \times C$	**$225,000**	**$300,000**
Non-marketing profit margin	m		50%	50%
Non-marketing costs	n	$m \times T$	$112,500	$150,000
Marketing costs	c_T	$v \times c$	$100,000	$100,000
Total profit	P	$T - (n + c_T)$	**$12,500**	**$50,000**
Total marketing ROI	R	$P \div c_T$	**13%**	**50%**

Source: Brian Clifton, Advanced Web Metrics with Google Analytics

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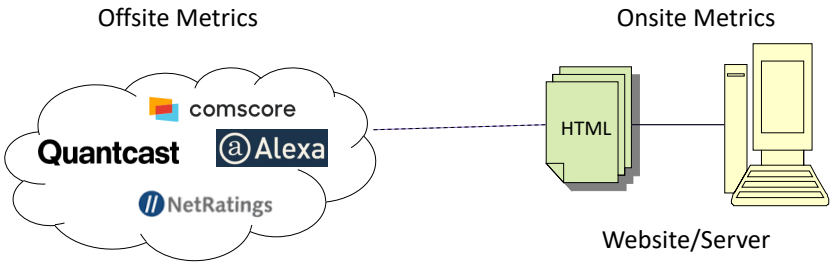
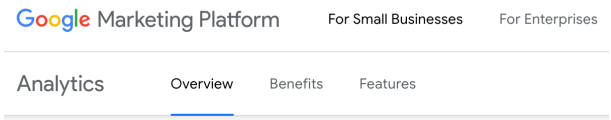
Available Data





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From UA (Universal Analytics) to GA4 (Google Analytics 4)

GA4 is event-first and designed for cross-platform journeys (web + apps).

Implication for marketing:

- The "source of truth" is no longer a complete clickstream.
- Expect missing data, more modeling, and greater reliance on experiments and first-party data.

https://support.google.com/analytics/answer/10089681?hl=en
https://support.google.com/analytics/answer/12159447?hl=en&ref_topic=14089939

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Signals Getting Noisier

Three forces reshaping digital marketing:

Privacy + consent

More users decline tracking; regulations and platform rules have changed.

Effect: fewer observable events, heavier reliance on consent-aware tagging and modeling.

Platform fragmentation

Walled gardens (social, retail media, marketplaces) keep data inside their ecosystems.

Effect: attribution gaps, reporting based on platform-provided metrics.

AI-mediated discovery

Search and social discovery increasingly happen inside AI-driven interfaces.

Effect: new journeys, new ad placements, different measurement assumptions.

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Privacy & Security Regulations

Recent regulations on privacy and what they changed for marketing analytics.

Timeline

- 2018 - GDPR enforceable (EU): consent + user rights, large penalties.
- 2020 - CCPA effective (California): notice, access/delete, opt-out of "sale/share".
- 2020 - Schrems II: Privacy Shield invalidated, cross-border transfers need stronger safeguards.
- 2020-2021 - LGPD (Brazil) + PIPL (China): global privacy rules expand, transfer/localization constraints.
- 2023 - CPRA + U.S. state wave (VA/CO/CT): sensitive data rules and targeted-ads opt-outs.
- 2024 - EU DSA applicable + NIS2 transposition deadline: platform risk management and stronger cybersecurity duties.

Impact on digital marketing & analytics

- More consent gating → fewer observable sessions/events; greater reliance on modeled/aggregated reporting.
- First-party data becomes a strategic asset (logins, CRM joins, server-side tagging).
- Smaller retargeting audiences → more contextual.
- Heavier vendor/transfer governance (DPAs, SCCs, retention limits, localization).
- Security expectations rise across the marketing stack (least-privilege access, incident response, breach readiness).

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Third-Party Cookies

Third-party cookies are small text files created by websites other than the one a user is currently visiting, which are stored on the user's device and can be accessed by the third-party website to track user activity across different sites.
https://stape.io/blog/third-party-cookies

These cookies are most commonly introduced via embedded content, such as advertisements, analytics scripts, or social media widgets, hosted on the third-party's servers and integrated into the primary website's code.

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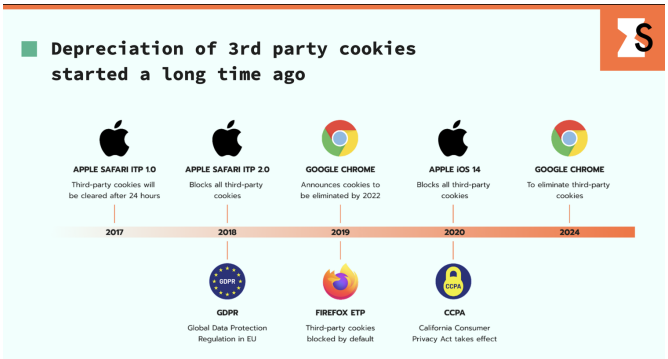
Third-Party Cookies: Current State



https://stape.io/blog/third-party-cookies

- Expect uneven cookie availability across browsers, and user settings.
- Invest in first-party data (auth, CRM) and robust UTMs.
- Use incrementality tests to validate channel impact when tracking is partial.

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Data in Modern Analytics

Observed
Directly measured (clicks, purchases)

Pros: concrete
Cons: incomplete

Modeled
Estimated via rules or statistical models

Pros: fills gaps
Cons: assumptions

Experimental
Incrementality via controlled tests

Pros: causal
Cons: slower/costly

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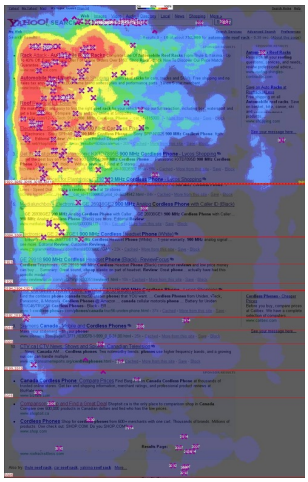
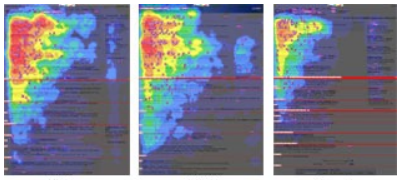
Eye-Tracking Studies



- Golden Triangle
  - Top left corner
  - Not only in search engines
- Quick scan
  - For candidate
- Longer scan
  - For relevance



Yahoo MSN Google

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More Recent Eye-Tracking for Google

https://www.forbes.com/sites/roberthof/2015/03/03/how-do-you-google-new-eye-tracking-study-reveals-huge-changes

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Clickstream / Log Analysis

- Eye-tracking studies are limited to a lab.
- Often, we need to analyze how users behave when visiting our site.
- Click data
  - Javascript tags on server
  - Cookies on browser
  - Google Analytics Tracking Code (GATC) for GA4
    - "hits" to Google Analytics

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Clickstream data from Similarweb

Engagement

Total Visits	2.26B ↓15.88%
Avg. Visit Duration	00:06:47
Pages per Visit	8.31
Bounce Rate	36.63%

Data source: SimilarWeb

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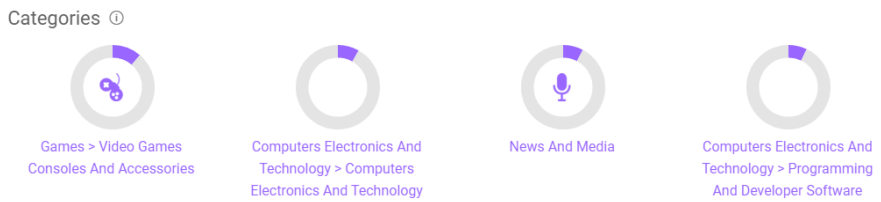
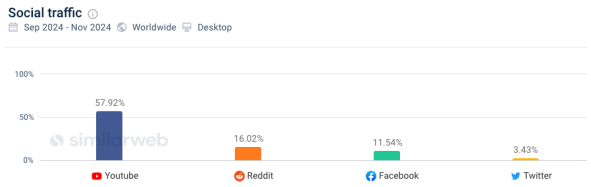
Audience Interests, Traffic Flow and more!





Some more measures …

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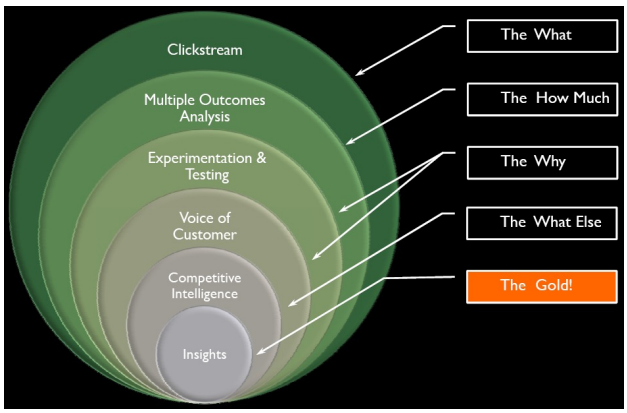
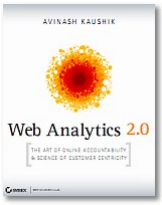
Web Analytics Framework

- Avinash Kaushik , analytics evangelist, formerly at Google





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What We Do?

- Multiple Outcomes Analysis (The How Much)
  - Increase revenue
  - Reduce cost
  - Improve customer satisfaction/loyalty
- Experimentation and Testing (The Why)
  - HiPPOs vs. customers
  - Experiment or die!
- Voice of Customer (The Why)
  - Surveys, lab usability testing, remote usability testing, and more
- Competitive intelligence (The What Else)

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Analytics 3.0 Framework Powered by AI?

- Conversational AI
  - Allows you to ask data questions in plain English.
- Prescriptive AI
  - AI tells you what to do next.

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Some Foundational Metrics

- Visitors / Unique visitors
  - Pay attention to definition of “unique” (cookie? date? IP?)
- Time on site (average session duration)
  - Tricky! Should consider the goal of the site
- Page views
  - Good for content/brand sites
  - Unclear for other sites
- Bounce rate (1 – engagement rate)
  - Reveals **real** visitors
  - An **engaged** visit in GA4 is any session that either:
    - Had 2 or more page views
    - Had 1 or more conversion events
    - Lasted 10 or more seconds

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Event-first measurement (GA4 mindset)

Define what happens, not just what page someone visited.

Event name
purchase
add_to_cart
lead_submit

Parameters
value
currency
item_id
channel

User properties
plan
region
account_type
cohort

A good event is…

- tied to a user action you care about such as purchase and sign_up (not a UI detail)
- consistent across web + app
- accompanied by parameters that answer “who/what/how much/where”
- stable over time (so cohorts and trend lines mean something)

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Content Metrics

- Top content
  - Why users are coming
  - What they are looking for
- Top landing (entry) pages
  - First impression!
  - Polish and direct users to goals
- Click density analysis
  - Use tools like CrazyEgg.com
- Funnel analysis
  - In multi-page processes, where users abandon?

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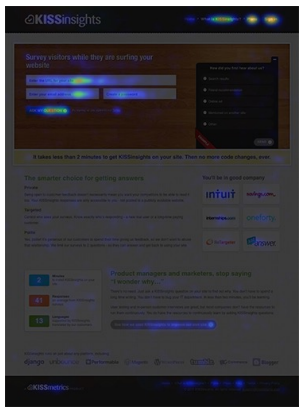
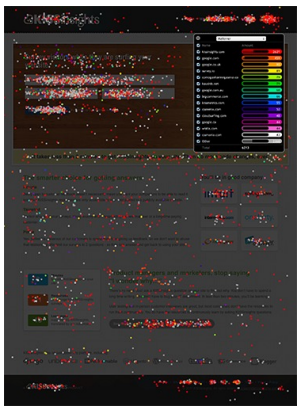
Click Density Analysis

Where users click, and which users click on each link

- Click Heatmap, Overlay

crazyegg. visualize your visitors





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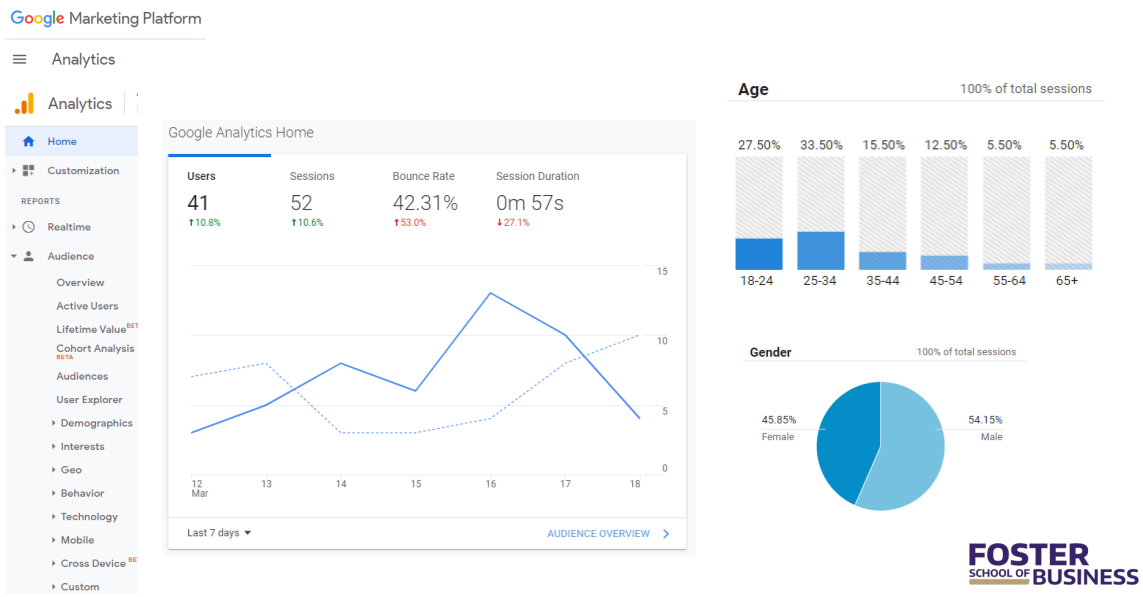
Google Analytics



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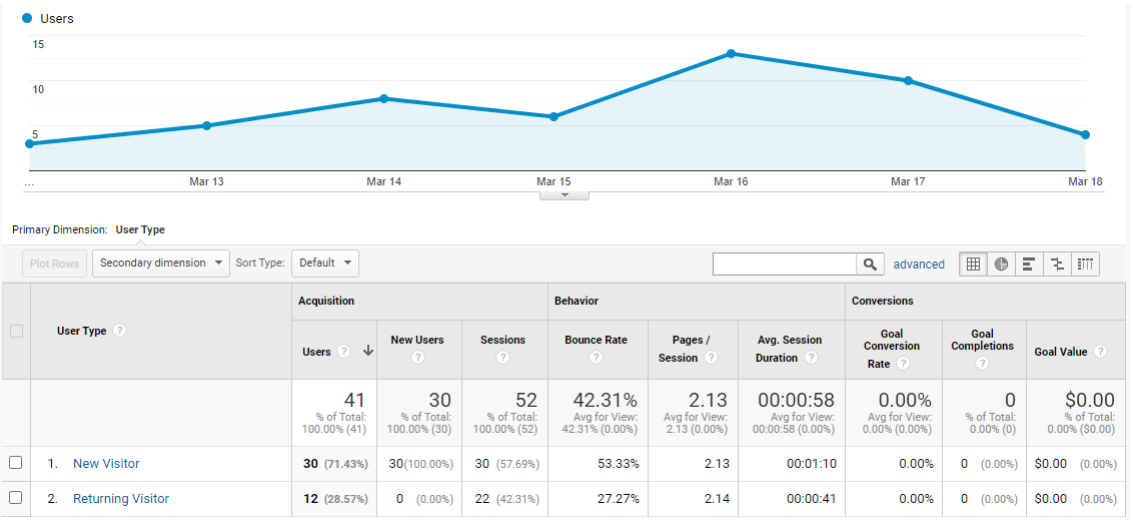
New vs. Returning



User Type	Acquisition			Behavior			Conversions		
	Users	New Users	Sessions	Bounce Rate	Pages / Session	Avg. Session Duration	Goal Conversion Rate	Goal Completions	Goal Value
	41 % of Total: 100.00% (41)	30 % of Total: 100.00% (30)	52 % of Total: 100.00% (52)	42.31% Avg for View: 42.31% (0.00%)	2.13 Avg for View: 2.13 (0.00%)	00:00:58 Avg for View: 00:00:58 (0.00%)	0.00% Avg for View: 0.00% (0.00%)	0 % of Total: 0.00% (0)	$0.00 % of Total: 0.00% ($0.00)
1. New Visitor	30 (71.43%)	30(100.00%)	30 (57.69%)	53.33%	2.13	00:01:10	0.00%	0 (0.00%)	$0.00 (0.00%)
2. Returning Visitor	12 (28.57%)	0 (0.00%)	22 (42.31%)	27.27%	2.14	00:00:41	0.00%	0 (0.00%)	$0.00 (0.00%)

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Engagement, Frequency and Recency

Sessions	Pageviews
52	111
% of Total: 100.00% (52)	% of Total: 100.00% (111)

Session Duration Bucket	Sessions	Pageviews
0-10 seconds	34	48
11-30 seconds	6	18
31-60 seconds	3	12
61-180 seconds	4	15
181-600 seconds	5	18

Count of Sessions	Sessions	Pageviews
1	30	64
2	5	9
3	1	2
4	1	3
5	2	3
6	1	5
7	1	4
8	2	3
15-25	4	7
26-50	5	11

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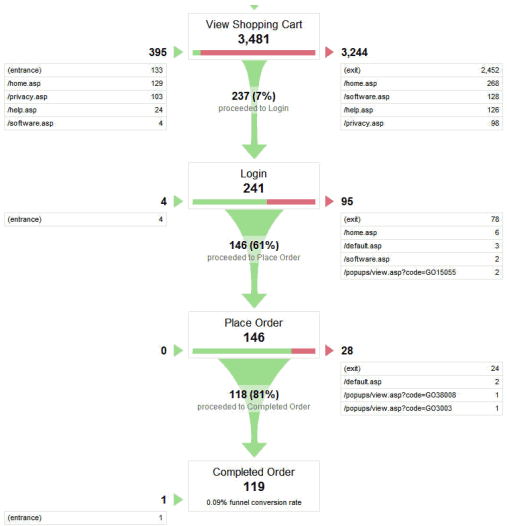
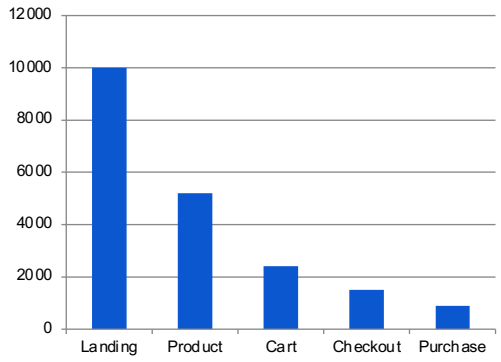
Marketing Funnel





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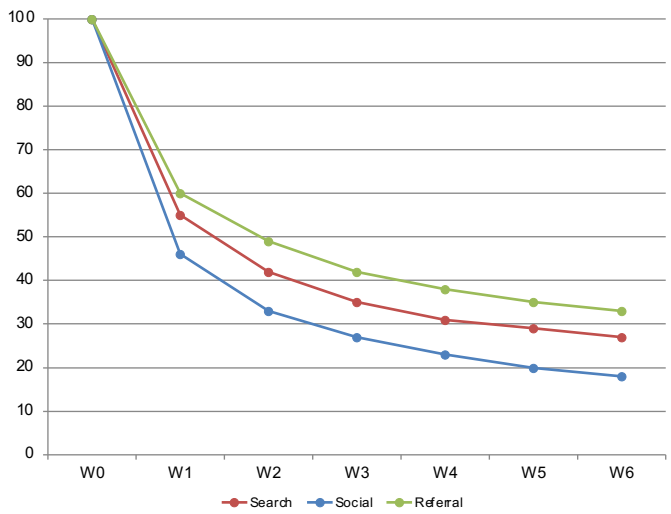
Cohorts



- Which channel brings users who stick?
- Is drop-off immediate (onboarding) or gradual (value)?
- Are you measuring retention in a meaningful way? (return visit, active use, repeat purchase)
- What message/offer drove the best cohort?

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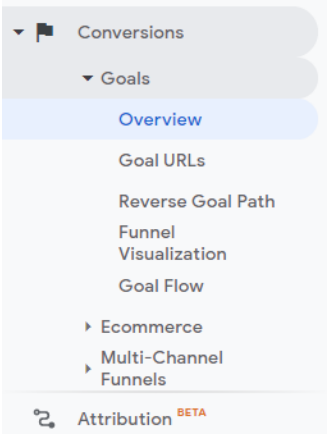
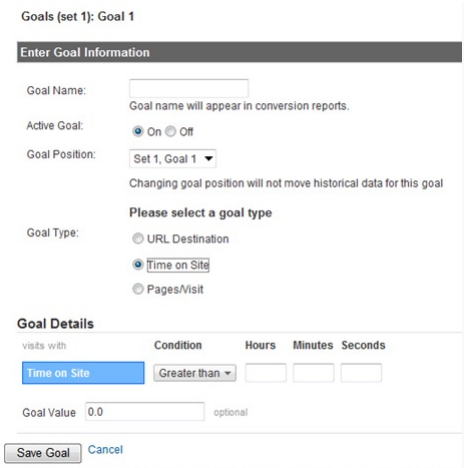
Goal and Lead Metrics

- Set up **goals** and measure **conversion rate** and **goal value**





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Attribution and Incrementality

Attribution

- Answers "who gets credit"?
- The inputs are the observed touchpoints and the output is attribution.
- Attribution approaches:
  - Last-touch
  - Data-driven

Approaches to measure incrementality

1. A/B test (on-site): change experience → measure conversion/retention lift
2. Holdout: suppress ads for a randomized group → compare outcomes
3. Geo test: turn ads on/off by region → measure lift (control for seasonality)
4. Matched markets: pair similar regions → improve signal

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Competitive Intelligence: Industry or Competitor Data

How customers behave in other websites (generate hypotheses about where users come from, where they go next, and why)

- Macro / Across-site / Competitive Intelligence
  - Panel data (ComScore, Alexa, SimilarWeb)
  - Search engine data (Google Trends)

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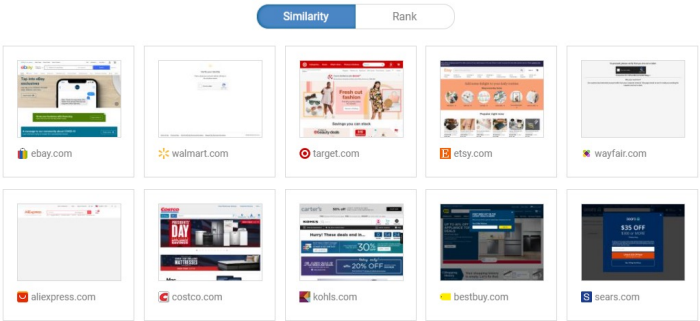
Need for Competitive Data

- Understand how competitors perform
- How competitors get visitors
- Where customers go after visiting competitor's site
- Demographics



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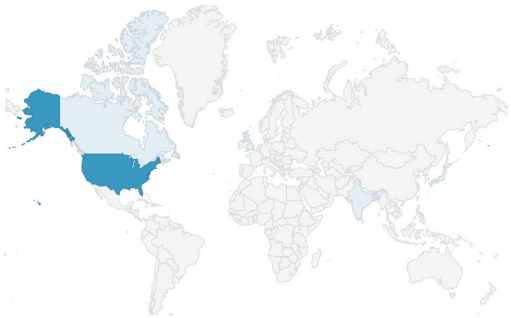
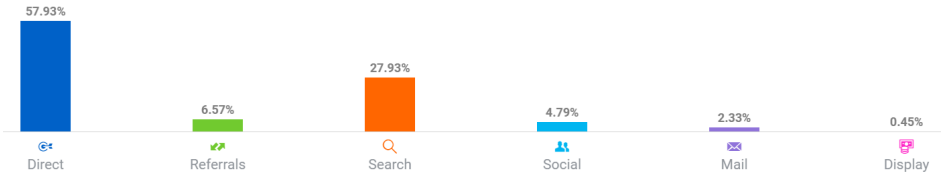
Sources of Traffic



United States	81.26%	16.00%
Canada	1.42%	15.12%
United Kingdom	1.24%	13.55%
India	1.15%	8.59%
Japan	0.73%	7.04%

See 247 more countries



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Search & Social

Search

27.93%
Of traffic is from Search

90.32% Organic

Top 5 Organic keywords
Out of 17,534:

amazon	25.79% ↓16.52%
amazon prime	6.40% ↓25.02%
amazon.com	1.50% ↓18.65%
amazon prime video	1.31% ↓5.36%
prime gaming	1.01% ↓9.15%

9.68% Paid

Top 5 Paid keywords
Out of 663:

amazon	7.48% ↓13.58%
amazon prime video	0.63% ↑3.14%
amazon prime	0.48% ↓11.75%
prime video	0.42% ↑1.98%
aws	0.19% ↓4.86%

Social

4.79%
Of traffic is from Social

Youtube	60.13%
Facebook	14.50%
Reddit	11.02%
Twitter	6.33%
Pinterest	3.58%

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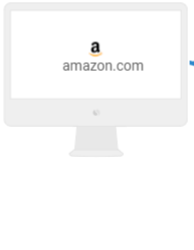
Upstream and Downstream

Top Referring Sites:

primevideo.com	22.61% ↓16.70%
audible.com	4.88% ↓13.53%
slickdeals.net	2.05% ↓21.01%
yahoo.com	1.64% ↑15.82%
dealnews.com	1.63% ↓21.32%



Top Destination Sites:

mail.google.com	8.41% ↓18.26%
primevideo.com	6.68% ↓20.52%
twitter.com	4.78% ↓18.68%
images-na.ssl-im...	4.69% ↓18.13%
amazon.syf.com	4.34% ↓20.51%

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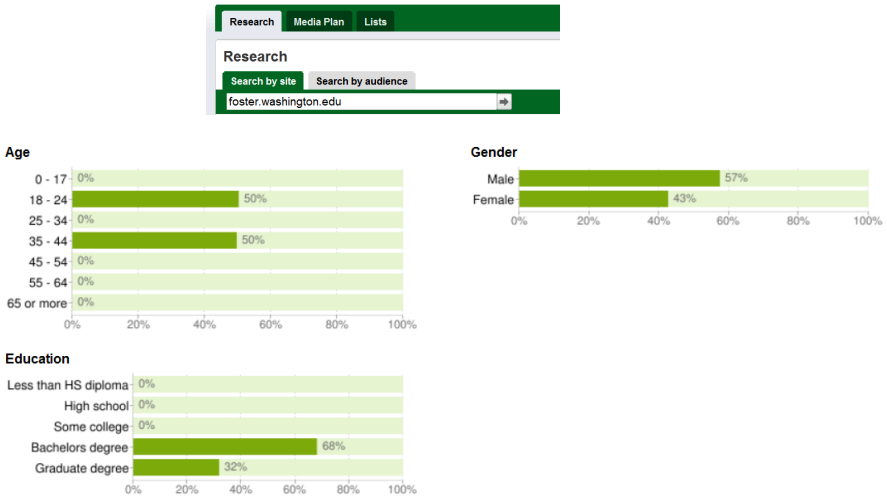
Demographics



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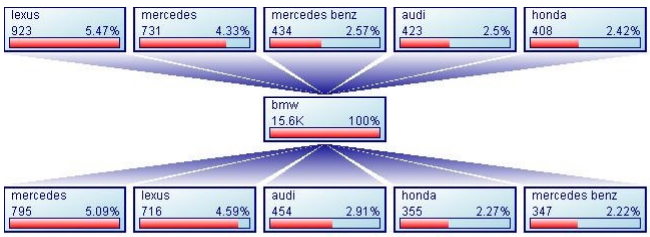
Analyzing Keyword Queries: Funneling

- What is the intent of customers that type such queries?
- Hint: What they searched before/after?
  - Search Funnels:



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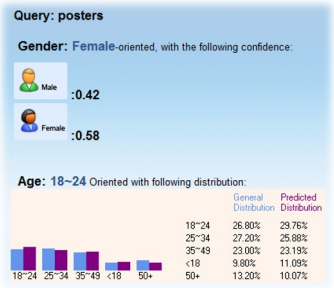
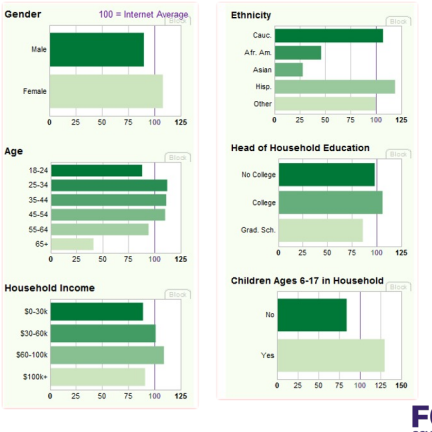
Keyword Demographics

- Keywords have demographic signatures
- Example: [posters]





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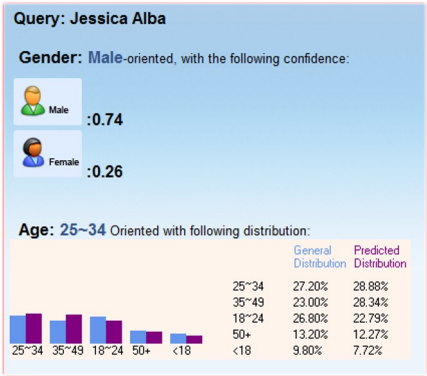
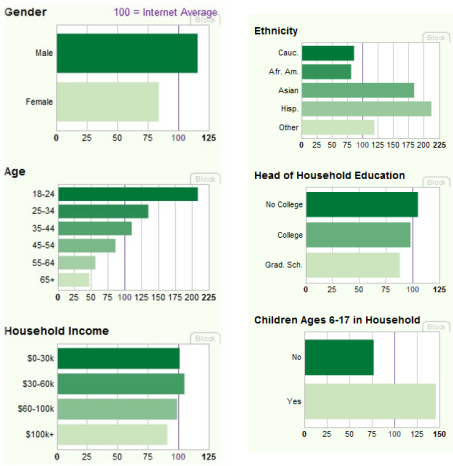
Keyword Demographics

- Example: [Jessica Alba]





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Going Beyond Existing Traffic

- So, we know what people come to the website.
- How can we get customers for **other keywords**?
  - Internal Search
  - Search Engine Optimization (SEO)
  - Pay-per-Click Advertising
  - Advertising Networks

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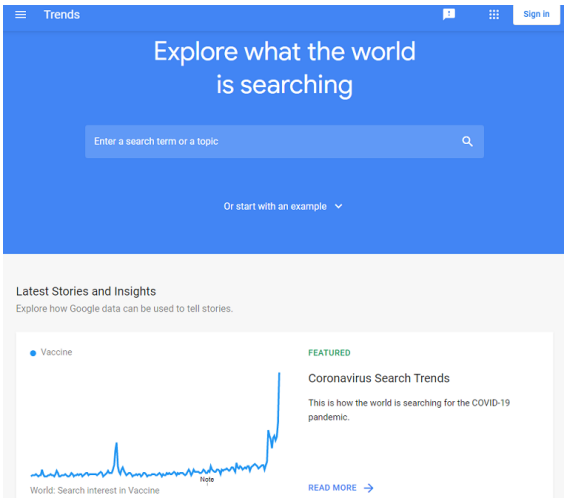
Forecasting with Google Trends

- What are people searching for on the web?
- Google Trends: www.google.com/trends



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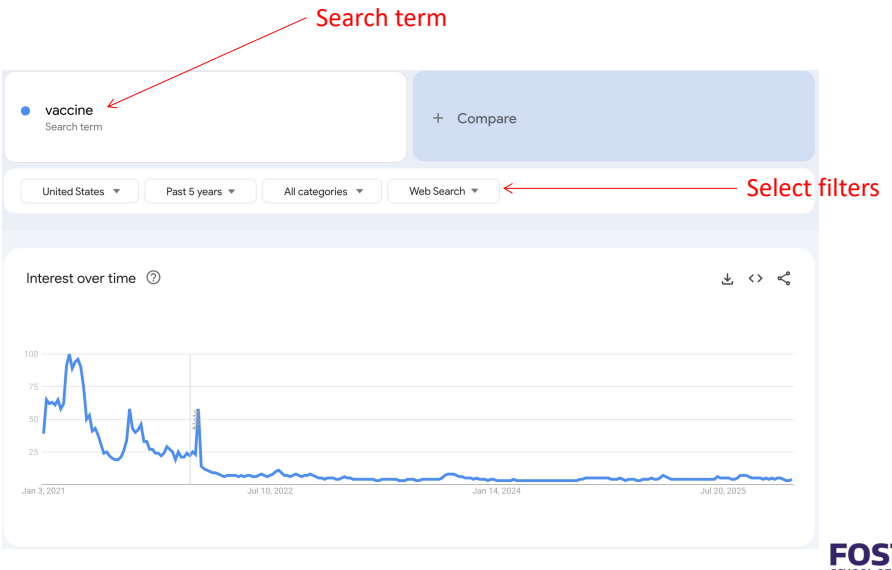
Google Trends



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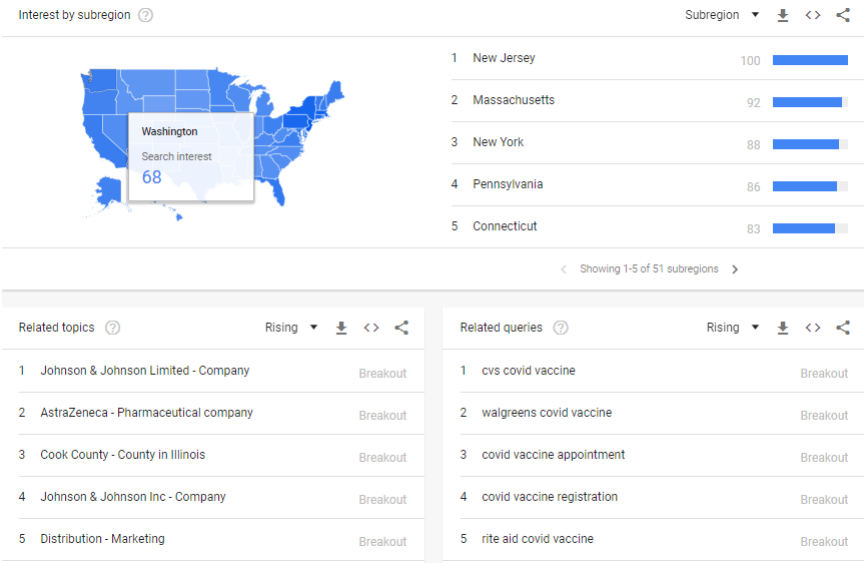
Google Trends



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What Are People Searching For?

Can this info help you in forecasting?

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Building a Predictor for US Housing Prices

- Search on Real Estate
- Search on all Subcategories of Real Estate like rental listings and mortgage listings

Predicting the present with Google Trends
H Choi, H Varian - Economic record, 2012

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Forecasting Primer

Basic forecasting models

- **Autoregressive**: value at time t depends on
  - Value at time t - 1
- **Seasonal adjustment**: value at time t depends on
  - Value at time t - 12
  - For monthly data
- **Transfer function**: value at time t depends on
  - Other contemporaneous or lagging variables
- Seasonal autoregressive transfer model: Value at time t depends on
  - Value at time t - 12 (seasonality)
  - Value at time t - 1 (recent behaviour)
  - Other lagging or contemporaneous variables (such as Google Trends data)
- Typical question of interest
  - How much more accurate forecasts can you get from additional variables over and above the accuracy you get with the history of the time series itself?

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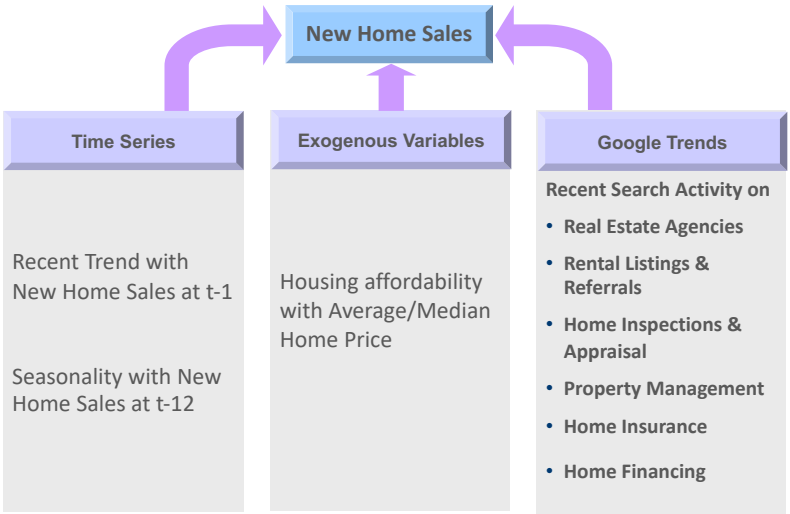
Model



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Predicting the Present: nowcasting

New Residential Sales from US Census

- Monthly release 24 – 28 days after the month
- Seasonally adjusted
- National and Regional aggregate

Google Trends Real Estate by Category

- Home Inspections & Appraisal
- Home Insurance
- Home Financing
- Property Management
- Rental Listings & Referrals
- Real Estate Agencies

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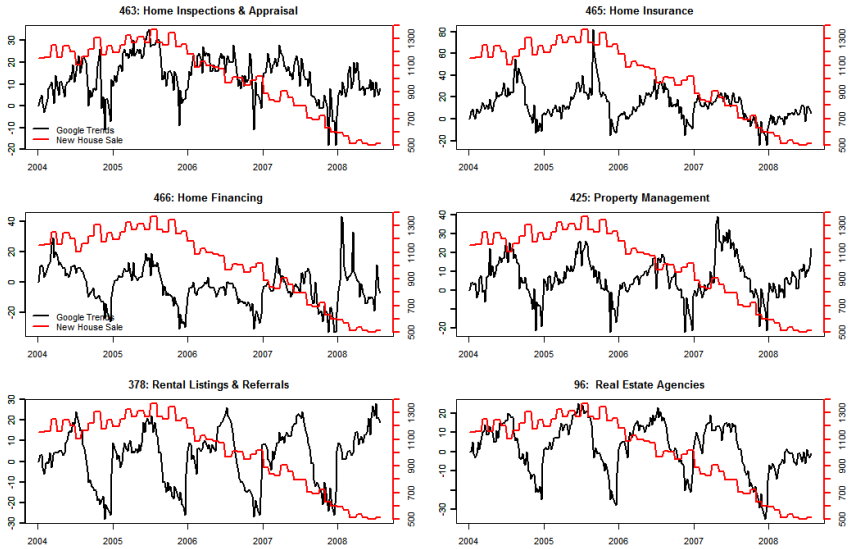
New House Sales vs. Real Estate Google Trends



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Analysis and Forecasting

Model:

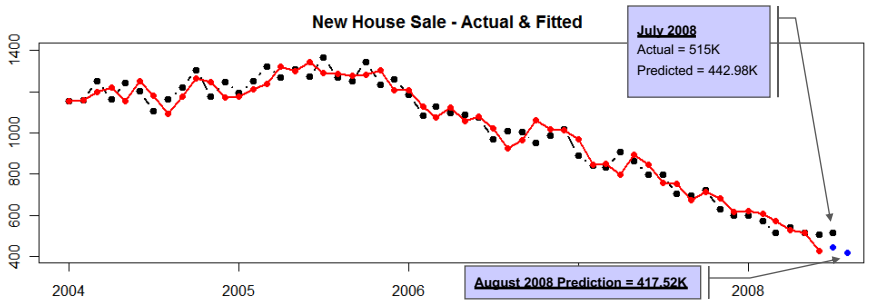
$Y_t = 446.1 + 0.864 * Y_{t-1} - 4.340 * us378.1 + 4.198 * us96.2 - 0.001 * AvgP_{t-1}$

Y_t : New house sold at month t

$AvgP_{t-1}$: Average Sales Price of New One-Family Houses Sold at (t-1)-th month

us378.1 : Google Trend of vertical id = 378 (Rental Listings & Referrals) at t-th month 1st week

us96.2 : Google Trend of vertical id = 96 (Real Estate Agent) at t-th month 2nd week



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Time Series Data – A Primer

- Forecast future values of a time series
- Four components of time series:
  - Level (average value of data over time)
  - Trend (overall direction of the data over time)
  - Seasonality (recurring patterns at regular intervals)
  - Noise (fluctuations we can't explain using level, trend and seasonality)

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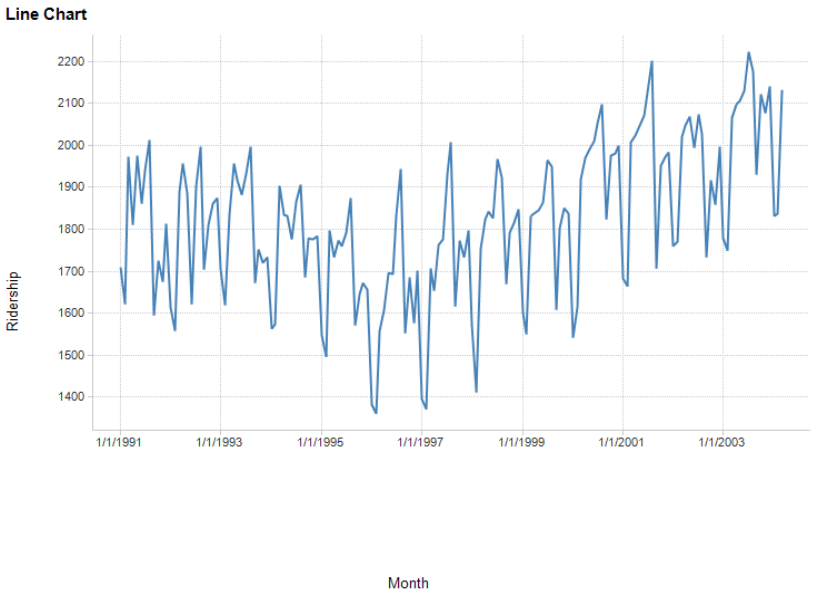
Amtrak Ridership



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Partitioning

- Divide data into training portion and validation portion
- Test model on the validation portion
- **Random partitioning** would leave holes in the data, which causes problems
  - Forecasting methods assume regular sequential data
- Instead of random selection, divide data into two parts
  - Train on early data
  - Validate on later data

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Regression-based Model

- Ridership Y is a function of time (t) and noise (error = e)

$$Y = B_0 + B_1 \times t + e$$

- Thus we model 3 of the 4 components:
  - Level (B_0)
  - Trend (B_1)
  - Noise (e)

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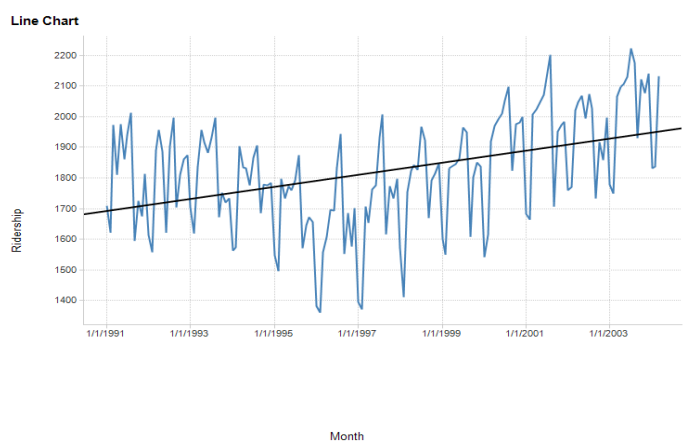
Model Fit

- The linear trend model is not a good fit
- Modify and use also for non-linear trends
  - Exponential
  - Polynomial
- Can also capture seasonality



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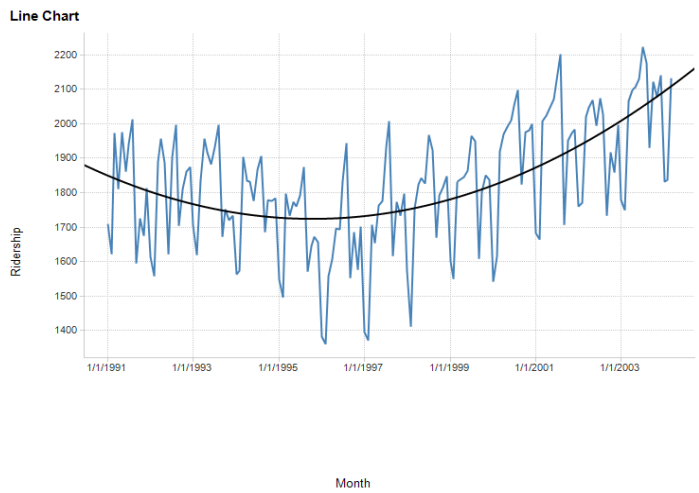
Polynomial Trend

- Add additional predictors as appropriate
- For example, for quadratic relationship add a t^2 predictor
- Fit linear regression using both t and t^2
- Higher order terms can be added too



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Handling Seasonality

- Seasonality is any recurring cyclical pattern of consistently higher or lower values (daily, weekly, monthly, quarterly, etc.)
- Handle in regression by adding categorical variables for season, e.g.,

Month	Ridership	Season
Jan-91	1709	Jan
Feb-91	1621	Feb
Mar-91	1973	March
Apr-91	1812	April

- Transform categorical variables into dummies
- To avoid multicollinearity, use $m - 1$ dummies for m categories

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